

# ISIT312/ISIT912 Big Data Management

## Subject Outline

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School of Computing and Information Technology -  
University of Wollongong

# ISIT312/912 Big Data Management

## Outline

### Lecturers

Subject organization

Assessment

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# Lecturers

Lecturer: Dr Behnaz Soltani

Email: [behnazs@uow.edu.au](mailto:behnazs@uow.edu.au)

Phone number: 42 395199

Office: Building 3, room 112

Consultation time:

- Monday 13.30 - 15.30
- Tuesday 12.00 - 14.00

Lecturer: Dr Janusz R. Getta

Email: [jrg@uow.edu.au](mailto:jrg@uow.edu.au)

Phone number: 42 214339

Office: Building 3, room 210

Consultation time:

- Monday 8.30 - 10.30
- Wednesday 12.30 - 14.30

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Session: **Spring 2026**

Credit points: **6**

Lectures: **13\*2 hours**

Laboratories: **11\*2 hours**

- Tuesday 12.30 - 14.30 in 3.126 at Wollongong Campus
- Tuesday 14.30 - 16.30 in 3.126 at Wollongong Campus
- Tuesday 16.30 - 18.30 in 3.126 at Wollongong Campus
- Tuesday 18.30 - 20.30 in 3.126 at Wollongong Campus
- Thursday 8.30 - 10.30 in 3.126 at Wollongong Campus
- Thursday 10.30 - 12.30 in 3.126 at Wollongong Campus
- Friday 14.30 - 16.30pm in 3.126 at Wollongong Campus
- Monday 13.30pm - 14.30pm in LP\_400-413 at Liverpool Campus

**There are no laboratory classes in week 1 and week 13 !**

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# Assessment

Three **assignments**, worth **50%** (10%+20%+20%) of the total evaluation in the subject

**Final examination** worth **50%** of the total evaluation in the subject

**Final examination** will consist of the problems similar to the problems included in the assignments

To be eligible to Pass this subject a student must achieve in the final examination a mark of at least 40% of the total evaluation of the final examination. Failure to meet one of the requirement will result in a TF (Technical Fail) grade being given if the overall subject mark is 50% or higher

All submissions of coursework (assignments) **MUST** be done through [Moodle site](#)

No submission will be accepted through email and/or in a paper form !

The subject has an in-class and paper-based final examination

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# Contents

BigData, Cluster Computing, Introduction to Hadoop

Hadoop Distributed Storage System (HDFS) and MapReduce Data Processing Model

MapReduce applications

Hive

Conceptual and logical Data Warehouse Design

SQL for Data Warehousing

Physical Data Warehouse design, Extract, Transform and Load (ETL)

HBase

Apache Pig

Spark fundamentals and data processing

Streaming data and advanced Spark operations

Very short introduction to Apache Kafka

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# Software

Virtual machine `BigData-Ubuntu22.04.4-Hadoop3.4.0-P3-Spark3.5.6` with [Hadoop](#), [Snakebite](#) and [Spark](#)

Virtual machine `BigData-Ubuntu22.04.4-Hadoop3.4.0-Hive4.0.1.ova` with [Hadoop](#) and [Hive](#)

Virtual machine `BigData-Ubuntu22.04.4-Hadoop3.4.0-Hbase2.6.1-hdfs.ova` with [Hadoop](#) and [HBase](#)

Virtual machine `BigData-Ubuntu22.04.4-Hadoop3.4.0-Pig0.17.0.ova` with [Hadoop](#) and [Pig](#)

The links to `ova` images of virtual machines are available on Moodle in a section [Resources](#)

Please use Chrome browser when downloading the `ova` files !

# Software

Conceptual modeling tool **UMLet 15.1**

A link to **UMLet 15.1** is available on Moodle in a section **Resources**

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# Subject outline

All important information related to the subject is included in the **Subject outline** available on **Moodle**

You **MUST** read the **Subject outline** for **ISIT312/ISIT912 Big Data Mangement !**

Do you have any questions ?